The Quantitative Single-Set Selection Principle for Sharp Faber–Krahn Stability (Route A)

Abstract

This note is the building block SelectionPrinciple of the Route A program for sharp Faber–Krahn stability with exponent 1/2,

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/N}\big(\lambda_1(\Omega)-\lambda_1(B^*)\big)\geq c_{\mathrm{FK}}(N)\,\mathcal{A}(\Omega)^2.$$

We isolate one quantitative step of the Brasco–De Philippis–Velichkov proof [1] (BDV): given one near-counterexample Ω_0 with small Saint–Venant quotient, we produce a *single* selected quasi-open set $\widetilde{\Omega}$ of unit volume whose energy-to-asymmetry quotient is controlled by that of Ω_0, on which the BDV free-boundary regularity package applies, and which carries an explicit volume-rescaled Bernoulli law on $\partial\widetilde{\Omega}$.

The proof contains no compactness or contradiction step. All BDV results are quoted as quantitative black boxes whose constants depend only on N and R (and are extractable from BDV, but not made numerically explicit here).

1 Setting and Normalisation

Fix $N\geq 2$ and $R\geq 2$. We use the BDV torsion convention: for a quasi-open set $\Omega\subset B_R\subset\mathbb{R}^N$,

$$E(\Omega):=\min_{u\in H^1_0(\Omega)}\Big\{\tfrac12\int_\Omega|\nabla u|^2-\int_\Omega u\Big\}=-\tfrac12\int_\Omega u_\Omega,$$

where u_Ω is the torsion function. The Saint–Venant deficit is

$$\delta(\Omega):=E(\Omega)-E(B_1)\geq 0\qquad\text{whenever } |\Omega|=|B_1|=\omega_N.$$

Asymmetries.

Let $\mathcal{A}(\Omega)$ be the Fraenkel asymmetry. Following BDV we work with the smoother asymmetry

$$\alpha(\Omega):=\int_{\Omega\Delta B_1(x_\Omega)}\big|1-|x-x_\Omega|\big|\,dx,\qquad x_\Omega:=\frac{1}{|\Omega|}\int_\Omega x\,dx.$$

BDV Lemma 4.2 gives, for $\Omega\subset B_R$,

$$|\Omega\Delta B_1(x_\Omega)|^2\leq C_1(N)\,\alpha(\Omega),\tag{1}$$
$$|\alpha(\Omega_1)-\alpha(\Omega_2)|\leq C_2(R)\,|\Omega_1\Delta\Omega_2|.\tag{2}$$

Throughout we set, with a small abuse of notation matching downstream blocks,

$$A(\Omega):=\alpha(\Omega),\qquad Q_\alpha(\Omega):=\frac{E(\Omega)-E(B_1)}{\alpha(\Omega)}.$$

Penalised functional.

Let $f_{\hat{\eta}}$ be the BDV one-sided volume penalty from Lemma 4.5,

$$f_{\hat{\eta}}(s) = \begin{cases} \hat{\eta}(s-\omega_N), & s \le \omega_N, \\ \hat{\eta}^{-1}(s-\omega_N), & s \ge \omega_N, \end{cases}$$

with $\hat{\eta} = \hat{\eta}(N,R) \in (0,1)$. Set $F_{\hat{\eta}}(U) := E(U) + f_{\hat{\eta}}(|U|)$. BDV Lemma 4.5 gives that B_1 minimises $F_{\hat{\eta}}$ over quasi-open $U \subset B_R$, and the linear bilateral bound

$$F_{\hat{\eta}}(U) - F_{\hat{\eta}}(B_1) \ge C_4(N,R)^{-1}\big||U| - |B_1|\big| \tag{3}$$

holds.

Given an input $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$, write $\varepsilon := \alpha(\Omega_0)$, $\delta := E(\Omega_0) - E(B_1)$, $q := \delta/\varepsilon$, and choose $\tau := q^{1/4}$. Define

$$J(U) := F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}. \tag{4}$$

This is BDV's penalised functional (4.10) with the BDV sequence parameter σ replaced by a single-set parameter τ tied to the input deficit.

Thresholds.

All depending only on N, R:

- $\tau_{\mathrm{ex}}(N,R) \le \hat{\eta}/(2C_2)$, ensuring existence of a minimiser of J and the uniform perimeter bound of BDV Lemma 4.6;
- $\sigma_2(N,R)$ from BDV Lemma 4.9 (density estimates);
- $\sigma_3(N,R)$ from BDV Lemma 4.16 (smooth Bernoulli coefficient);
- $\tau_{\mathrm{reg}}(N,R) := \min\{\tau_{\mathrm{ex}}, \sigma_2, \sigma_3\}$.

We assume

$$q \le q_{\mathrm{sel}}(N,R) := \big(\tau_{\mathrm{reg}}(N,R)\big)^4, \qquad \varepsilon \le \varepsilon_{\mathrm{sel}}(N,R). \tag{5}$$

The second smallness keeps the dilated set inside a fixed enlarged ambient ball. Under (5), $\tau = q^{1/4} \le \tau_{\mathrm{reg}}$.

2 Existence by the BDV Direct Method

Lemma 1 (Existence + uniform perimeter; BDV Lemma 4.6, single-set form)**.** *For every $\tau \le \tau_{\mathrm{ex}}(N,R)$, J admits a quasi-open minimiser $U_* \subset B_R$, $|U_*| > 0$, with*

$$P(U_*) \le P_{\mathrm{sel}}(N,R).$$

Why this is not a hidden compactness step. The BDV existence proof for (4.10) applies the direct method to the truncated lower-semicontinuous part $U \mapsto F_{\hat{\eta}}(U)$, using

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2} \ge \tau\,|\alpha(U) - \varepsilon|$$

together with (2) to bound $\alpha(U)$ along a minimising sequence and force $|U| \le M(N,R)$. The closure step uses γ-convergence of capacitary measures inside B_R as in BDV §2, whose constants depend only on N, R. Importantly, the limit set is admitted as a minimiser without any selection from a sequence of *counterexamples*: the input here is a single Ω_0 and existence is for the functional J at fixed parameters (ε, τ).

The uniform perimeter bound follows from BDV (4.13)–(4.14): testing $J(U_*) \le J(\Omega_0) \le F_{\hat{\eta}}(\Omega_0) + \varepsilon$ and using the isoperimetric inequality with (3) produces a bound depending only on N, R. □

We fix henceforth a minimiser U_* from Lemma 1.

3 The Three Quantitative Lemmas

Lemma 1. Energy and asymmetry preservation before normalisation.

Lemma 2. *The minimiser U_* satisfies*

$$0 \le F_{\hat\eta}(U_*) - F_{\hat\eta}(B_1) \le \delta, \tag{6}$$

$$|\alpha(U_*) - \varepsilon| \le \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}. \tag{7}$$

In particular, with $\tau = q^{1/4}$ and $q \le 1$, $|\alpha(U_*) - \varepsilon| \le 2\varepsilon\, q^{1/4}$.

Proof. By minimality $J(U_*) \le J(\Omega_0) = F_{\hat\eta}(\Omega_0) + \varepsilon$. Since $|\Omega_0| = |B_1|$, $F_{\hat\eta}(\Omega_0) = E(\Omega_0) = F_{\hat\eta}(B_1) + \delta$. As B_1 minimises $F_{\hat\eta}$, $F_{\hat\eta}(U_*) \ge F_{\hat\eta}(B_1)$. Hence

$$F_{\hat\eta}(U_*) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \le F_{\hat\eta}(B_1) + \delta + \varepsilon,$$

giving (6) and $\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \le \varepsilon + \delta$. Squaring yields (7). □

Lemma 2. Volume error.

Lemma 3. *There is $C_5 = C_5(N, R)$ such that*

$$\big||U_*| - |B_1|\big| \le C_5\,\delta, \qquad |r_* - 1| \le C_5\,\delta, \quad r_* := (|U_*|/|B_1|)^{1/N}.$$

Proof. By Schwarz symmetrisation, $F_{\hat\eta}(B_{r_*}) \le F_{\hat\eta}(U_*)$ where B_{r_*} is a ball of volume $|U_*|$. Combining with (3) and (6),

$$C_4^{-1}\big||U_*| - |B_1|\big| \le F_{\hat\eta}(B_{r_*}) - F_{\hat\eta}(B_1) \le F_{\hat\eta}(U_*) - F_{\hat\eta}(B_1) \le \delta.$$

Hence $||U_*| - |B_1|| \le C_4\delta$, and bilipschitz dependence of r_* on $|U_*|$ near 1 yields $|r_* - 1| \le C_5\delta$. □

Lemma 3 (fully expanded). Quotient preservation after volume normalisation.

Define $\widetilde\Omega := r_*^{-1} U_*$, $r_* = (|U_*|/|B_1|)^{1/N}$.

Lemma 4. *Under* (5), *with $\tau = q^{1/4}$, after translation by $-x_{U_*}$ if needed:*

(a) $|\widetilde\Omega| = |B_1| = \omega_N$.

(b) $|\alpha(\widetilde\Omega) - \varepsilon| \le \rho\,\varepsilon$ *with* $\rho = 2q^{1/4} + C_\alpha(N, R)\,q \le 1/2$.

(c) $E(\widetilde\Omega) - E(B_1) \le C_E(N, R)\,\delta$.

(d) $Q_\alpha(\widetilde\Omega) \le \dfrac{C_E(N, R)}{1 - \rho}\, Q_\alpha(\Omega_0)$.

The proof rests on three quantitative inputs.

Step 1: Lipschitz dependence of α under dilation. Set $\lambda := r_*^{-1}$. The symmetric difference $U_*\Delta\lambda U_*$ is contained in an annulus of width $|\lambda-1|R$; by the BDV perimeter bound,

$$|U_*\Delta\lambda U_*| \le R\,P(U_*)\,|\lambda-1| + o(|\lambda-1|) \le 2RP_{\mathrm{sel}}(N,R)\,|\lambda-1|, \tag{8}$$

for $|\lambda-1|\le 1/2$. Combined with (2),

$$|\alpha(\widetilde{\Omega}) - \alpha(U_*)| \le C_2(R)\,|U_*\Delta\lambda U_*| \le 2RC_2P_{\mathrm{sel}}\,|\lambda-1|.$$

By Lemma 3, $|\lambda-1|\le 2C_5\delta$ when $|r_*-1|\le 1/2$. Hence

$$|\alpha(\widetilde{\Omega}) - \alpha(U_*)| \le C_\alpha(N,R)\,\delta, \qquad C_\alpha := 4RC_2P_{\mathrm{sel}}C_5.$$

Combined with Lemma 2,

$$|\alpha(\widetilde{\Omega}) - \varepsilon| \le |\alpha(\widetilde{\Omega}) - \alpha(U_*)| + |\alpha(U_*) - \varepsilon| \le \varepsilon\Big(C_\alpha q + \frac{\sqrt{2q+q^2}}{\tau}\Big) = \rho\varepsilon. \tag{9}$$

With $\tau = q^{1/4}$ and small q, $\rho \le 2q^{1/4} + C_\alpha q \le 1/2$.

Step 2: Energy scaling. For quasi-open $V\subset B_R$ and $\mu>0$ with $\mu V\subset B_R$,

$$E(\mu V) = \mu^{N+2}E(V). \tag{10}$$

Indeed, if u_V is the torsion function of V, then $u_{\mu V}(x) = \mu^2 u_V(x/\mu)$ minimises $\frac{1}{2}\int|\nabla\cdot|^2 - \int\cdot$ on $H_0^1(\mu V)$, and the change of variable gives (10).

Step 3: From $E(U_*)$ to $E(\widetilde{\Omega})$. Apply (10) with $\mu=\lambda=r_*^{-1}$:

$$E(\widetilde{\Omega}) - E(B_1) = \lambda^{N+2}\big(E(U_*) - E(B_1)\big) + (\lambda^{N+2}-1)E(B_1).$$

From Lemma 2 and (3),

$$E(U_*) - E(B_1) \le \delta + \hat{\eta}^{-1}\big||U_*| - |B_1|\big| \le (1+\hat{\eta}^{-1}C_4)\delta =: C_6\delta.$$

Since $|\lambda-1|\le 1/2$, $\lambda^{N+2}\le(3/2)^{N+2} =: C_7(N)$ and $|\lambda^{N+2}-1|\le C_8(N)C_5\delta$. Thus

$$E(\widetilde{\Omega}) - E(B_1) \le (C_6C_7 + C_8C_5|E(B_1)|)\,\delta =: C_E(N,R)\,\delta. \tag{11}$$

Step 4: Quotient. By (9) and (11),

$$Q_\alpha(\widetilde{\Omega}) = \frac{E(\widetilde{\Omega}) - E(B_1)}{\alpha(\widetilde{\Omega})} \le \frac{C_E\delta}{(1-\rho)\varepsilon} = \frac{C_E}{1-\rho}q = \frac{C_E}{1-\rho}Q_\alpha(\Omega_0).$$

This proves (d).

4 The Selection Theorem

Theorem 5 (Single-set quantitative selection)**.** *Fix $N\ge 2, R\ge 2$. Let $\alpha_{\mathrm{sel}}(N,R) := \varepsilon_{\mathrm{sel}}(N,R)$, $q_{\mathrm{sel}}(N,R) := \tau_{\mathrm{reg}}(N,R)^4$. Suppose $\Omega_0\subset B_R$ is quasi-open with $|\Omega_0| = |B_1|$, $0<\varepsilon := \alpha(\Omega_0) \le \alpha_{\mathrm{sel}}$, $q := Q_\alpha(\Omega_0) \le q_{\mathrm{sel}}$. Set $\tau := q^{1/4}$ and let U_* be a minimiser of J. Define $\widetilde{\Omega} := r_*^{-1}U_*$, $r_* = (|U_*|/|B_1|)^{1/N}$. Then there exist $\rho=\rho(q)\le 1/2$ and $C_{\mathrm{sel}}(N,R)>0$ such that:*

(i) $|\widetilde{\Omega}| = |B_1|$;

(ii) $(1-\rho)\varepsilon \le \alpha(\widetilde{\Omega}) \le (1+\rho)\varepsilon$, $\rho\le 1/2$;

(iii) $Q_\alpha(\widetilde{\Omega}) \leq \dfrac{C_{\mathrm{sel}}(N,R)}{1-\rho}\, q$;

(iv) $\widetilde{\Omega}$ *inherits the BDV penalised-minimiser regularity package (two-sided density estimates, Lipschitz/nondegeneracy of the torsion potential, smooth Bernoulli coefficient);*

(v) *the selected Bernoulli law on* $\partial\widetilde{\Omega}$ *takes the volume-rescaled form*

$$|\nabla \widetilde{u}(y)|^2 = \widetilde{\Lambda} + \widetilde{a}_\sigma\, \widetilde{\Psi}_{r_*}(y), \qquad \widetilde{\Lambda} = r_*^{-2}\Lambda, \quad \widetilde{a}_\sigma = r_*^{-1} a_\sigma.$$

Proof. Parts (i)–(iii) are Lemma 4. For (iv)–(v) see Sections 5–6. □

5 Free-Boundary Regularity Inherited from BDV

The minimiser $U_* = \{u_* > 0\}$ satisfies the perturbed one-phase quasi-minimality inequality (BDV (4.19) with σ replaced by τ): for every $v \in H_0^1(B_R)$,

$$\tfrac{1}{2}\int |\nabla u_*|^2 - \int u_* + f_{\hat{\eta}}(|\{u_* > 0\}|) \leq \tfrac{1}{2}\int |\nabla v|^2 - \int v + f_{\hat{\eta}}(|\{v > 0\}|) + C_R\, \tau\, \big|\{u_* > 0\}\Delta\{v > 0\}\big|, \tag{12}$$

the error being BDV's bound for the variation of $\sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}$, dominated by $\tau\, C_2(R)|U\Delta U_*|$.

For $\tau \leq \tau_{\mathrm{reg}}(N,R)$ the BDV Alt–Caffarelli regularity package applies with constants depending only on N, R (Lemmas 4.9, 4.12, 4.16). The volume-normalisation $\widetilde{\Omega} = r_*^{-1}U_*$ is a bilipschitz dilation with $|\lambda - 1| \leq 1/2$; all density, Lipschitz, nondegeneracy and C^k constants transfer with degradation at most $(3/2)^N$.

6 The Volume-Normalised Bernoulli Law

BDV Lemma 4.16 gives on ∂U_*,

$$|\nabla u_{U_*}(x)|^2 = \Lambda + a_\sigma\, \Psi_{U_*}(x), \qquad |a_\sigma| \leq \tau, \tag{13}$$

with $\Psi_{U_*}(x) = |x - x_{U_*}| - V_{U_*}\cdot x$ and $V_{U_*} = \int_{U_*} \frac{z - x_{U_*}}{|z - x_{U_*}|}\, dz$.

Lemma 6 (Rescaled Bernoulli law). *Set* $\widetilde{u}(y) := r_*^{-2} u_{U_*}(r_* y)$. *Then* $-\Delta\widetilde{u} = 1$ *in* $\widetilde{\Omega}$, $\widetilde{u} = 0$ *on* $\partial\widetilde{\Omega}$, *and on* $\partial\widetilde{\Omega}$

$$|\nabla\widetilde{u}(y)|^2 = \widetilde{\Lambda} + \widetilde{a}_\sigma\, \widetilde{\Psi}_{r_*}(y), \tag{14}$$

with $\widetilde{\Lambda} = r_*^{-2}\Lambda$, $\widetilde{a}_\sigma = r_*^{-1}a_\sigma$, *and*

$$\widetilde{\Psi}_{r_*}(y) := |y - r_*^{-1}x_{U_*}| - r_*^N V_{\widetilde{\Omega}}\cdot y.$$

This differs from the unscaled α-bracket only by the harmless factor $r_*^N = 1 + O(\delta)$ in the barycentric term.

Proof. $\nabla\widetilde{u}(y) = r_*^{-1}\nabla u_{U_*}(r_* y)$ and $\Delta\widetilde{u}(y) = \Delta u_{U_*}(r_* y) = -1$. On $\partial\widetilde{\Omega}$, $r_* y \in \partial U_*$, so by (13), $|\nabla\widetilde{u}(y)|^2 = r_*^{-2}\Lambda + r_*^{-2}a_\sigma\, \Psi_{U_*}(r_* y)$. Change of variables in the centroid integral gives $V_{U_*} = r_*^N V_{\widetilde{\Omega}}$ and $\Psi_{U_*}(r_* y) = r_*\big[|y - r_*^{-1}x_{U_*}| - r_*^N V_{\widetilde{\Omega}}\cdot y\big]$. Reorganising yields (14). Since $|r_* - 1| \leq C_5\delta$ both prefactors degrade by $1 + O(\delta)$. □

7 Summary of Constants

Symbol	Source	Dependence
$\hat{\eta}, C_4$	BDV Lemma 4.5	N, R
C_1	BDV Lemma 4.2(i)	N
C_2	BDV Lemma 4.2(ii)	R
σ_2, σ_3	BDV Lemmas 4.9, 4.16	N, R
$\tau_{\text{ex}}, \tau_{\text{reg}}$	this note	N, R
P_{sel}	BDV Lemma 4.6	N, R
C_5	Lemma 3	N, R
$C_\alpha = 4RC_2P_{\text{sel}}C_5$	Step 1 of Lemma 4	N, R
C_E	Step 3 of Lemma 4	N, R
$C_{\text{sel}} = C_E$	final quotient	N, R
$\varepsilon_{\text{sel}}, q_{\text{sel}}$	smallness regime (5)	N, R
$\rho(q) = 2q^{1/4} + C_\alpha q$	(9)	$N, R,\ q$
$\widetilde{\Lambda} = r_*^{-2}\Lambda,\ \widetilde{a}_\sigma = r_*^{-1}a_\sigma$	rescaled Bernoulli	N, R

All constants depend only on (N, R) via named BDV constants. None is set by a compactness or contradiction step.

References

[1] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831 (arXiv:1306.0392).